K J SOMAIYA INSTITUTE OF TECHNOLOGY
Department of Information Technology

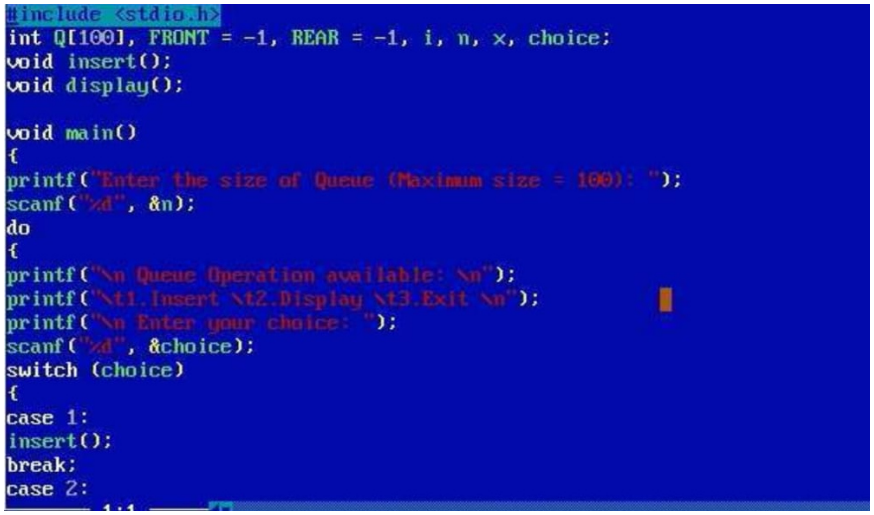
Academic Year: 2026-27
Class: S.Y. B.Tech.
Semester: III

Subject: DSA LAB
Course Code: ITL302

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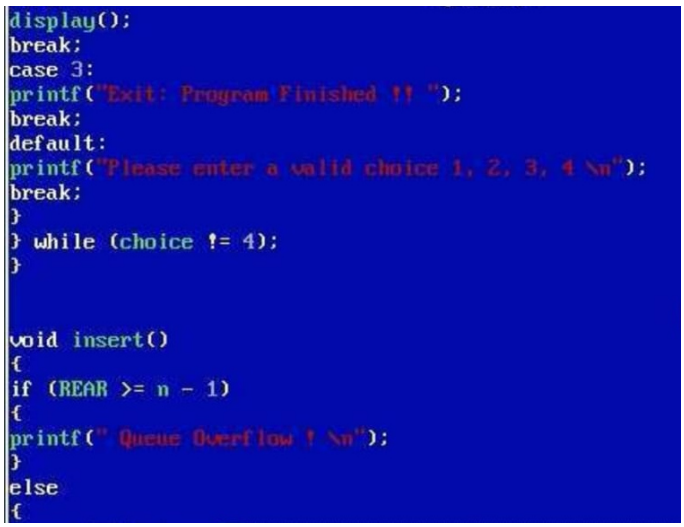
EXP NO:2

Program:



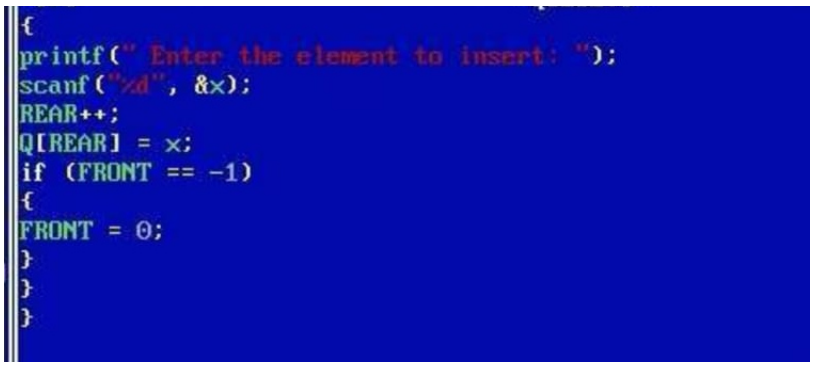
```
#include <stdio.h>
int Q[100], FRONT = -1, REAR = -1, i, n, x, choice;
void insert();
void display();

void main()
{
printf("Enter the size of Queue (Maximum size = 100): ");
scanf("%d", &n);
do
{
printf("\n Queue Operation available: \n");
printf("\t1.Insert \t2.Display \t3.Exit \n");
printf("\n Enter your choice: ");
scanf("%d", &choice);
switch (choice)
{
case 1:
insert();
break;
case 2:
```



```
display();
break;
case 3:
printf("Exit: Program Finished !! ");
break;
default:
printf("Please enter a valid choice 1, 2, 3, 4 \n");
break;
}
} while (choice != 4);
}


void insert()
{
if (REAR >= n - 1)
{
printf(" Queue Overflow ! \n");
}
else
{
```



```
{
printf(" Enter the element to insert: ");
scanf("%d", &x);
REAR++;
Q[REAR] = x;
if (FRONT == -1)
{
FRONT = 0;
}
}
}
```

```
void display()
{
    if (REAR < 0)
    {
        printf(" Queue is empty ! \n");
    }
    else
    {
        printf(" The elements in the Queue are: \n");
        for (i = FRONT; i < n; i++)
        {
            printf(" %d ", Q[i]);
        }
        printf("\n");
    }
}
```

Output:

```
Enter the size of Queue (Maximum size = 100): 2

Queue Operation available:
       1.Insert        2.Display       3.Exit

Enter your choice: 1
Enter the element to insert: 10

Queue Operation available:
       1.Insert        2.Display       3.Exit

Enter your choice: 1
Enter the element to insert: 20

Queue Operation available:
       1.Insert        2.Display       3.Exit

Enter your choice: 2
The elements in the Queue are:
10  20
```